Developer Tools

Developing apps around your data in AWS

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How can users access AWS ?

- To access AWS, you have three options:
  - **AWS Management Console** (protected by password + MFA)
  - **AWS Command Line Interface (CLI):** protected by access keys
  - **AWS Software Developer Kit (SDK)** - for code: protected by access keys
- Access Keys are generated through the AWS Console
- Users manage their own access keys
- **Access Keys are secret, just like a password. Don't share them**
- Access Key ID ~= username
- Secret Access Key ~= password

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Example (Fake) Access Keys

Access keys

Use access keys to make secure REST or HTTP Query protocol requests to AWS service APIs. For your protection, you should never share your secret keys with anyone. As a best practice, we recommend frequent key rotation. Learn more

Create access key

Access key ID	Created	Last used	Status	
AKIASK4E37PV4TU3RD6C	2020-05-25 15:13 UTC+0100	N/A	Active | Make inactive	✖

- Access key ID: AKIASK4E37PV4983d6C
- Secret Access Key: AZPN3zojWozWCndIjhB0Unh8239a1bzbzO5fqqkZq
- **Remember: don't share your access keys**

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What's the AWS CLI?

- A tool that enables you to interact with AWS services using commands in your command-line shell
- Direct access to the public APIs of AWS services
- You can develop scripts to manage your resources
- It's open-source https://github.com/aws/aws-cli
- Alternative to using AWS Management Console

```
➜  ~ aws s3 cp myfile.txt s3://ccp-mybucket/myfile.txt
upload: ./myfile.txt to s3://ccp-mybucket/myfile.txt
➜  ~ aws s3 ls s3://ccp-mybucket
2021-05-14 03:22:52          0 myfile.txt
➜  ~
```

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What's the AWS SDK?

- AWS Software Development Kit (AWS SDK)
- Language-specific APIs (set of libraries)
- Enables you to access and manage AWS services programmatically
- Embedded within your application
- Supports
  - SDKs (JavaScript, Python, PHP, .NET, Ruby, Java, Go, Node.js, C++)
  - Mobile SDKs (Android, iOS, …)
  - IoT Device SDKs (Embedded C, Arduino, …)
- Example: AWS CLI is built on AWS SDK for Python

Your Application

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AWS Cloud Development Kit (CDK)

- Define your cloud infrastructure using a familiar language:
  - JavaScript/TypeScript, Python, Java, and .NET
- Contains high level components called **constructs**
- The code is “compiled” into a CloudFormation template (JSON/YAML)
- **You can therefore deploy infrastructure and application runtime code together**
  - Great for Lambda functions
  - Great for Docker containers in ECS / EKS

```
export class MyEcsConstructStack extends core.Stack {
  constructor(scope: core.App, id: string, props?: core.StackProps)
    super(scope, id, props);

    const vpc = new ec2.Vpc(this, "MyVpc", {
      maxAzs: 3 // Default is all AZs in region
    });

    const cluster = new ecs.Cluster(this, "MyCluster", {
      vpc: vpc
    });

    // Create a load-balanced Fargate service and make it public
    new ecs_patterns.ApplicationLoadBalancedFargateService(this, "My
      cluster: cluster, // Required
      cpu: 512, // Default is 256
      desiredCount: 6, // Default is 1
      taskImageOptions: { image: ecs.ContainerImage.fromRegistry("am
      memoryLimitMiB: 2048, // Default is 512
      publicLoadBalancer: true // Default is false
    });
  }
}
```

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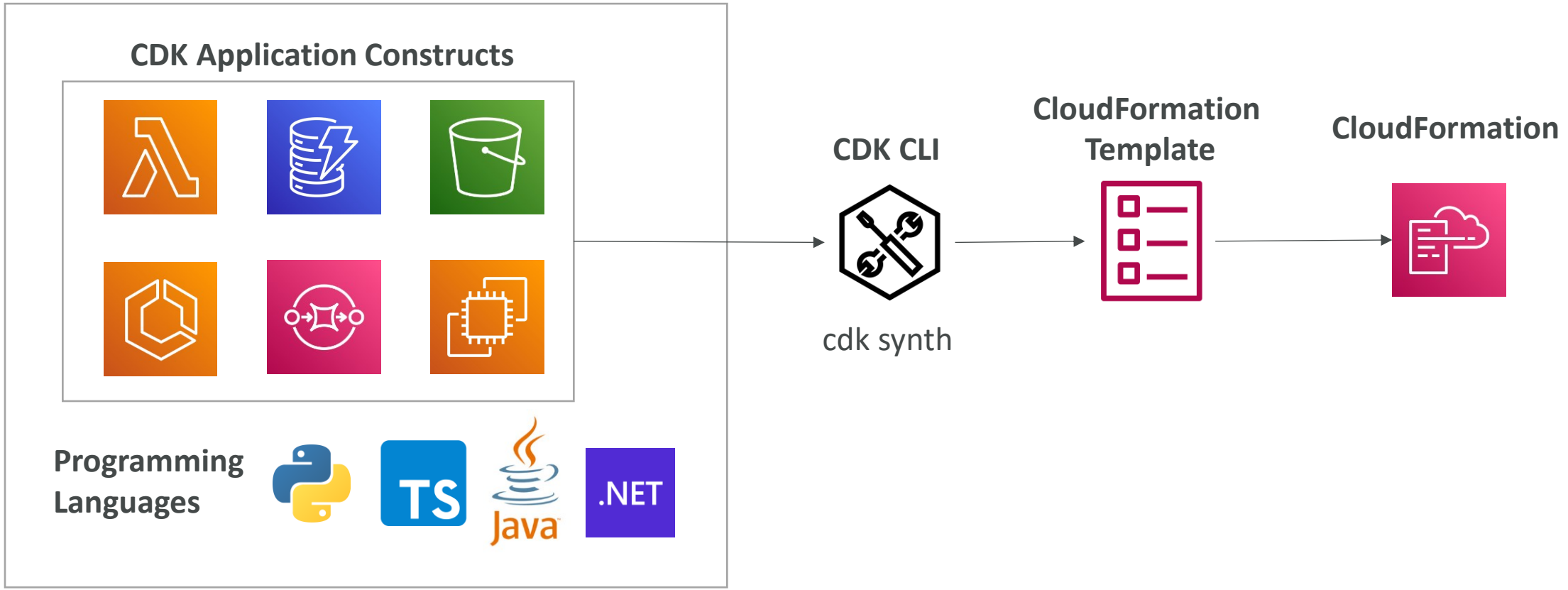
CDK in a diagram



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CDK vs SAM

- SAM:
  - Serverless focused
  - Write your template declaratively in JSON or YAML
  - Great for quickly getting started with Lambda
  - Leverages CloudFormation

- CDK:
  - All AWS services
  - Write infra in a programming language JavaScript/TypeScript, Python, Java, and .NET
  - Leverages CloudFormation

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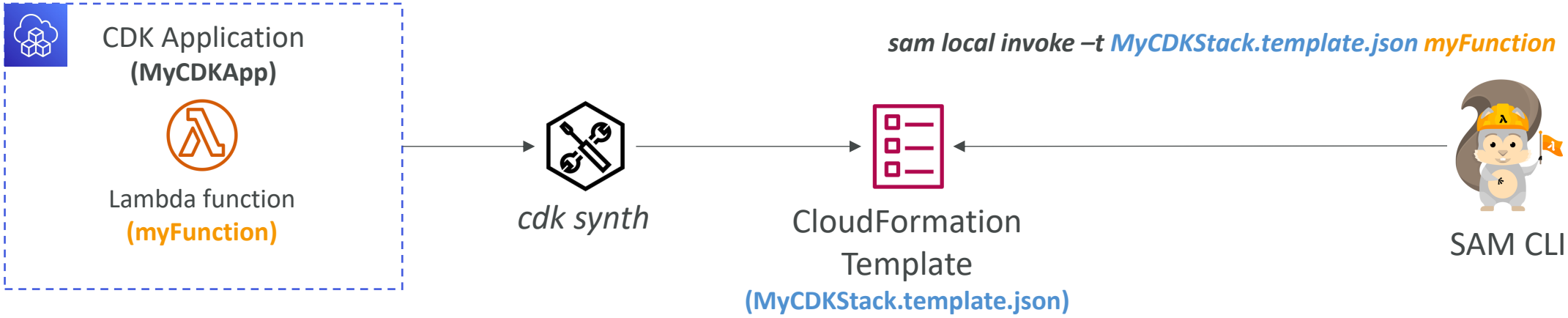
CDK + SAM

- You can use SAM CLI to locally test your CDK apps
- **You must first run *cdk synth***



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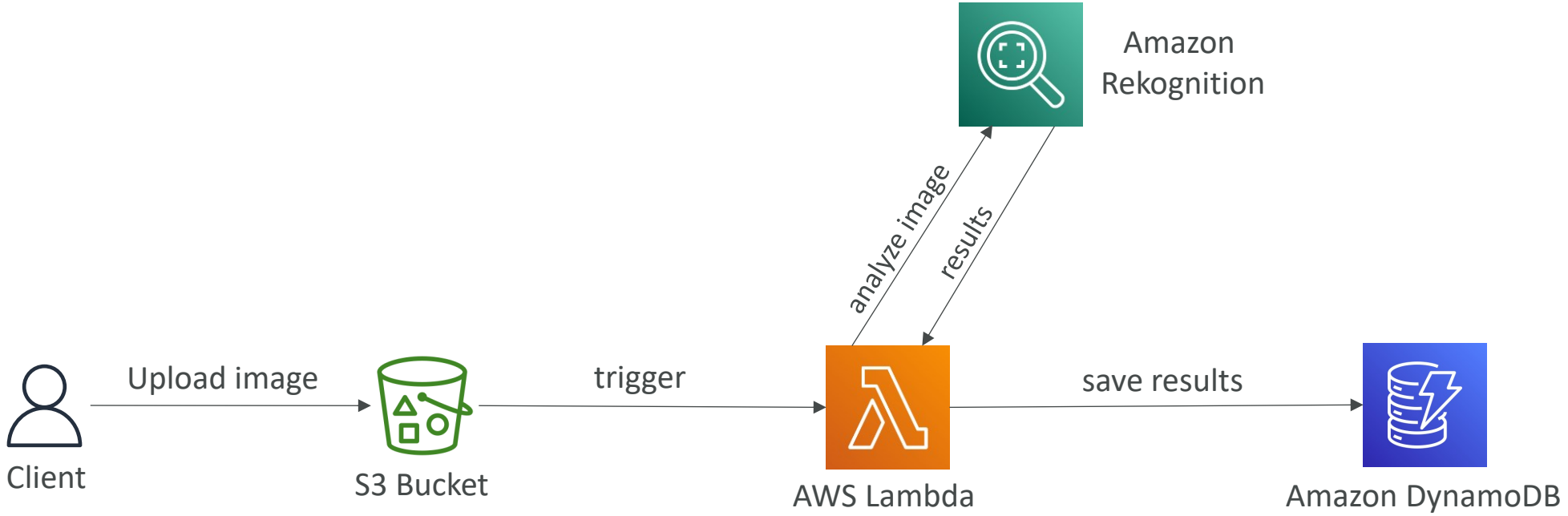
CDK Hands-On



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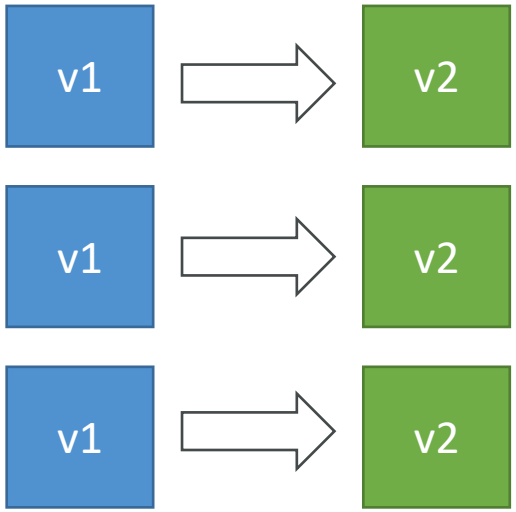
AWS CodeDeploy

- We want to deploy our application **automatically**

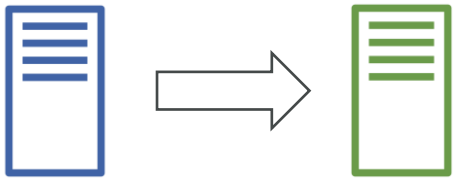
- **Works with EC2 Instances**
- **Works with On-Premises Servers**
- **<u>Hybrid</u> service**

- Servers / Instances must be provisioned and configured ahead of time with the CodeDeploy Agent

EC2 Instances being upgraded



On-premises Servers being upgraded



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CodeCommit – Important – Deprecation

- On July 25th 2024, AWS abruptly discontinued CodeCommit
- New customers cannot use the service
- AWS recommends to migrate to an external Git solution
- For this course:
  - CodeCommit might still appear at the exam (for now)
  - Every time I mention CodeCommit, assume there's a GitHub integration

GitHub GitLab git 3rd party

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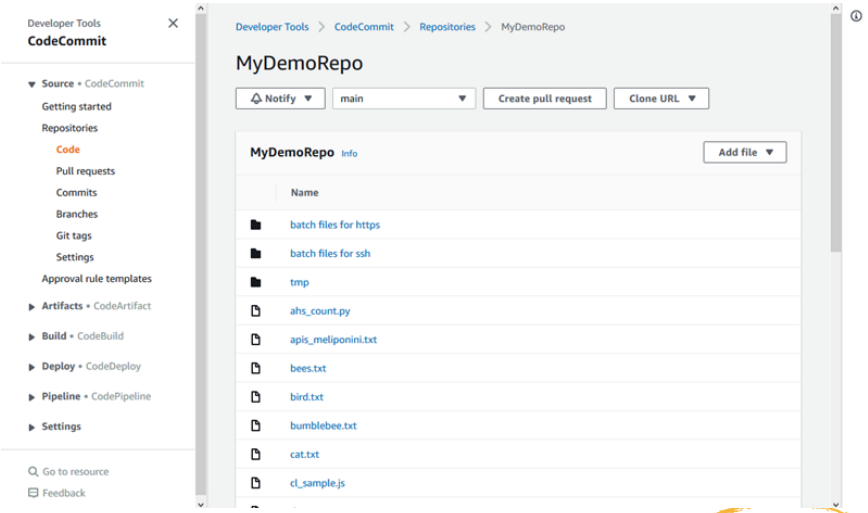
AWS CodeCommit

- Before pushing the application code to servers, it needs to be stored somewhere
- Developers usually store **code in a repository, using the Git technology**
- A famous public offering is GitHub, AWS' competing product is **CodeCommit**
- CodeCommit:
  - Source-control service that **hosts Git-based repositories**
  - Makes it easy to **collaborate with others on code**
  - The code changes are automatically **versioned**
- Benefits:
  - Fully managed
  - Scalable & highly available
  - Private, Secured, Integrated with AWS



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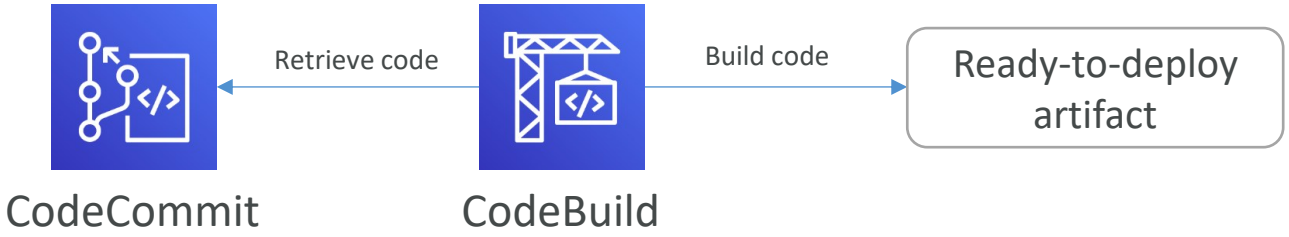
AWS CodeBuild

- Code building service in the cloud (name is obvious)
- **Compiles source code, run tests, and produces packages that are ready to be deployed (by CodeDeploy for example)**



- Benefits:
  - Fully managed, serverless
  - Continuously scalable & highly available
  - Secure
  - Pay-as-you-go pricing – only pay for the build time

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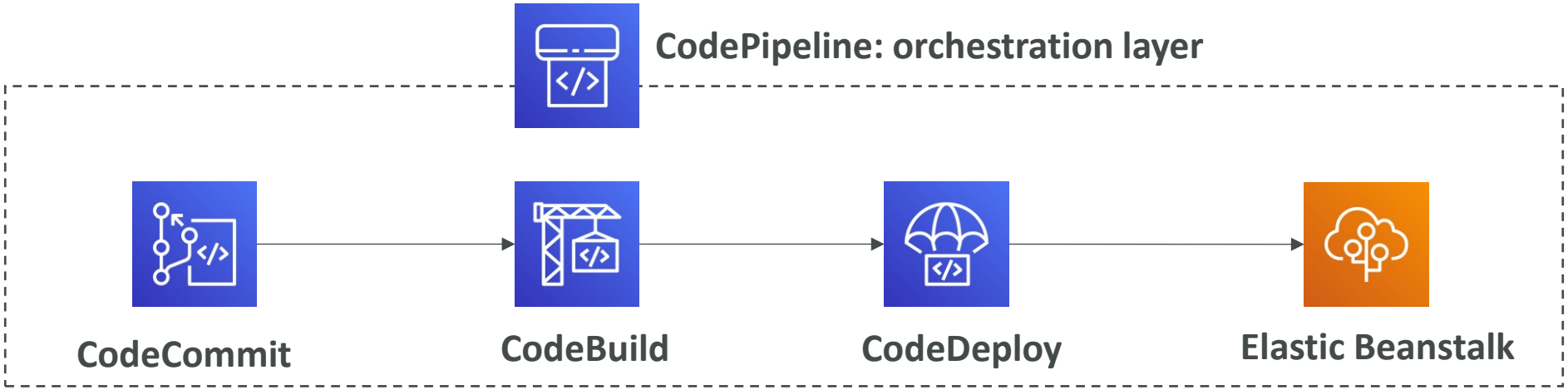
AWS CodePipeline

- **Orchestrate the different steps to have the code automatically pushed to production**
  - Code => Build => Test => Provision => Deploy
  - Basis for CICD (Continuous Integration & Continuous Delivery)
- Benefits:
  - Fully managed, compatible with CodeCommit, CodeBuild, CodeDeploy, Elastic Beanstalk, CloudFormation, GitHub, 3rd-party services (GitHub…) & custom plugins…
  - Fast delivery & rapid updates



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